

ML Based Trajectory Optimization for UAV Enabled 5G Communication

Raj Pohekar, Vineet Mundada, Atharva Sonar *, and Ankita Patel

Abstract: The Unmanned Aerial Vehicles (UAVs) are also a legitimate alternative by maximizing the coverage, capacity, and reliability of the fifth generation (5G) wireless communication systems due to their all-around movement, versatile deployment, and their ability to establish direct air-to-ground connections. UAV communication assistance would be particularly beneficial during the process of disaster recovery and events with the mass population, where the infrastructure on the ground may be unreachable or inundated. However, optimization of the UAV trajectories is challenging because time varies between wireless channels, insufficient onboard power, and user demands. In the paper, a machine learning-driven trajectory optimization model will be introduced, which is energy-constrained in UAV-aided 5G communication networks. The UAV issue of planning the route is formulated as a sequential decision-making problem and solved by the use of deep reinforcement learning, such that it becomes feasible to become adaptable and environment-conscious in mobility. The framework that will be offered will consist of probabilistic line-of-sight modeling, air-to-ground path loss modeling, signal-to-noise ratio modeling, achievable data rate modeling, and UAV energy consumption modeling. Results of the simulation show that the best UAV path is the one that enables a more effective data rate average at the cost of 18–25 percent and 20–30 percent of the total less energy by the UAV. The most preferred UAV relay flattens the mean signal power of the user from -40.8 dB to -32.6 dB (the gain of 8.2 dB). Furthermore, the likelihood of coverage increases from 0.46 to 0.81, and the index of fairness also increases to 0.88, which implies that there is a more balanced resourcing of signals among the users. These results suggest that reinforcement learning can be useful when deploying UAV-aided 5G in adaptive and energy-efficient modes.

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1 Introduction

Cellular networks tend to become unavailable or overloaded, especially during emergencies like disasters and big events, thus rendering communication difficult. Implementation of new infrastructure is usually not feasible [1, 4]. UAVs are an adaptable answer; they can serve as aerial base stations, provide temporary 5G connectivity by air-ground communication links, and can be deployed quickly [2, 4].

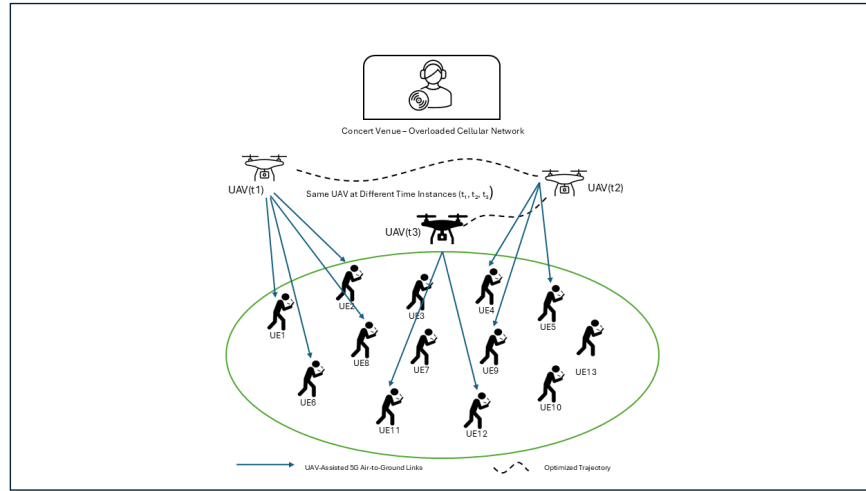


Fig. 1 UAV-enabled wireless communication scenarios

Fig. 1 shows that in a high-density event, a UAV is flying between time instances (t_1, t_2, t_3) on an optimized trajectory to enhance coverage and connectivity to ground users with the assistance of a 5G system [1, 18, 21]. The UAV trajectory is one of the factors that has a strong impact on the performance of UAV-assisted communication systems. Poor coverage, increased path loss, lower achievable data rates, and excessive energy consumption may be caused by an inefficient flight path, and this restricts the operating time of the UAV in severe cases [6, 18]. Conventional trajectory planning methods tend to be either fixed or heuristic and do not have the flexibility to adapt to dynamic network circumstances and time-varying user distributions [11, 20].

To overcome these obstacles, this paper presents a machine learning-based trajectory optimization system for 5G communication systems using UAVs. Reinforcement learning is utilized to allow adjustable and energy-efficient UAV movement by modeling UAV trajectory planning as a progressive choice problem. The proposed framework combines probabilistic line-of-sight modeling, air-to-ground path loss calculation, signal-to-noise ratio calculation, achievable data rate modeling, and UAV energy consumption modeling into a unified learning process [9, 15]. The UAV acquires an optimal three-dimensional flight plan through continuous interaction with the wireless environment, which helps achieve a balance between communication performance and energy usage. Consequently, the proposed solution can be effectively applied to temporary and emergency communication setups, where flexibility, efficiency, and fast network recovery are of paramount importance.

2 Related Work

Table 1 Summary of Research Reviewed (Part 1)

Authors and Year	Algorithm(s)	Findings/Outcomes
Mahmood et al. [1]-2020	Joint 3D Trajectory, Resource Optimization	Optimizing trajectory and resources in throughput; expensive to compute.
Sun et al. [2]-2022	Comprehensive Machine Learning (ML) Survey	ML techniques including scalability and latency issues.
Hashesh et al. [3]-2021	AI-Enabled Communication Frameworks	Discovers loopholes in B5G/6G integration and autonomous control.
Lins et al. [4]-2021	AI-Driven Mobility Enhancement	AI-based enhancement of 5G reliability in critical missions.
Gong et al. [5]-2020	Bayesian Optimization with Deep RL	Efficient and fast Bayesian-DRL optimization.
Afifi & Gadallah [6]-2020	Cellular Signal Feature ML	GPS-less navigation; signals susceptible to jitter.
Gopi et al. [7]-2021	ML-Assisted Adaptive Modulation	Adaptive modulation stabilizes B5G links.
Abubakar et al. [8]-2021	Energy Optimization Survey	Evaluation of traditional and ML energy optimization.
Kouhdaragh et al. [9] – 2020	Supervised and Reinforcement Learning	RAN Design: Explores supervised and reinforcement learning for 5G radio access networks.
Hussain et al. [10] – 2021	Geometric Feature-Based ML	Path Loss Prediction: Applies geometric ML to model mmWave channels accurately.
Wang & Gurosoy [11] – 2021	Decentralized Deep Reinforcement Learning	Collision-free UAV trajectories and constant connectivity maintenance.
Rasheed et al. [12] – 2022	LSTM-Based Conditional GAN	Provides a data-driven channel model for maritime UAV communication.
Kurunathan et al. [13] – 2022	Survey of ML for UAV Operations	Comprehensive overview of ML in UAV control and communication systems.

Table 2 Summary of Research Reviewed (Part 2)

Authors and Year	Algorithm(s)	Findings/Outcomes
Asim et al. [14] – 2022	Joint Trajectory and IRS Beamforming	Optimizes IRS beamforming and UAV trajectory simultaneously.
Qian et al. [15] – 2023	Double Deep Q-Network (DDQN)	Enhances physical-layer security through trajectory and power optimization.
Li et al. [16] – 2024	Model Predictive Control (MPC)	Secure MPC-based beamforming and robust trajectory optimization.
Yuan et al. [17] – 2022	Non-Convex Optimization	Maximizes secure energy efficiency using power and trajectory control.
Fonseca et al. [18] – 2022	Deep Reinforcement Learning (DRL)	Adaptive altitude optimization strategy for large-scale 5G UAV deployments.

3 System Model

This part explains the UAV-aided 5G communication framework, network structure, UAV movement, channel model, power consumption, and optimization using reinforcement learning.

3.1 Network Model

We consider a UAV-based 5G system that offers ad hoc coverage during disaster recovery and other large-scale events. One UAV is used as an aerial base station. Ground user equipments (UEs) are spread over a specific area randomly. The UAV exchanges messages with UEs through downlink air-to-ground channels, as well as having a stable backhaul connection with a macro base station. The UAV is operated in 3D space, and thus it can dynamically change its position to maximize communication performance.

3.2 UAV Mobility and Trajectory Model

The UAV path is sampled into time intervals of duration Δt . At time t , the position of the UAV is

$$\mathbf{q}(t) = [x(t), y(t), h(t)] \quad (1)$$

where $x(t)$ and $y(t)$ denote horizontal coordinates and $h(t)$ denotes altitude.

The UAV velocity is defined as:

$$v(t) = \frac{\|\mathbf{q}(t+1) - \mathbf{q}(t)\|}{\Delta t} \quad (2)$$

The UAV movement is limited by maximum velocity and altitude.

3.3 Air-to-Ground Channel Model

The ground-air channel model is based on the probabilistic LoS/NLoS model. The mean path loss is given as:

$$PL = P_{LoS}PL_{LoS} + (1 - P_{LoS})PL_{NLoS} \quad (3)$$

where P_{LoS} denotes the probability of LoS propagation.

The received SNR at each UE is:

$$SNR = \frac{P_{tx}}{PL \cdot N_0} \quad (4)$$

where P_{tx} is the UAV transmitting power and N_0 denotes noise power.

The achievable data rate follows Shannon's formula:

$$R = B \log_2(1 + SNR) \quad (5)$$

where B is Bandwidth

3.4 Energy Consumption Model

The propulsion energy per time step is assumed to be:

$$E_t = c_1 \|v_u(t)\|^2 + c_2 \quad (6)$$

where c_1 and c_2 represent the motion and hovering energy coefficients, respectively.

The total energy over trajectory T is:

$$E_{total} = \sum_{t=1}^T E(t) \quad (7)$$

The remaining energy is as follows:

$$E_{rem}(t) = E_{init} - \sum_{\tau=1}^t E_{\tau} \quad (8)$$

To enable energy-aware optimization, the reward is defined as:

$$r_t = R_t - \lambda E_t \quad (9)$$

where R_t denotes balancing data rate and λ is a weighting factor

A Double Deep Q-Network (DDQN) is used to solve the trajectory optimization, with the target Q-value as follows:

$$y_t = r_t + \gamma Q_{\text{target}} \left(s_{t+1}, \arg \max_{a'} Q_{\text{online}}(s_{t+1}, a') \right) \quad (10)$$

Finally, the energy efficiency is evaluated as

$$\eta_E = \frac{\sum_{t=1}^T R_t}{E_{\text{total}}} \quad (11)$$

representing the achieved data rate per unit energy.

3.5 Reinforcement Learning Interaction Model

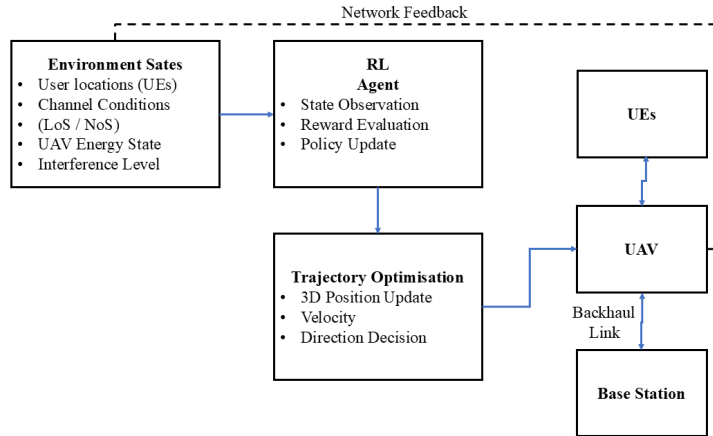


Fig. 2 System architecture of the RL-Based UAV trajectory optimization

The UAV trajectory optimization problem is developed as a sequence of decisions based on reinforcement learning, where the UAV is an intelligent agent that acts in a changing wireless environment. The environment includes the on-ground location of the user, air-ground channel conditions (LoS / NLoS), the interference, and the state of energy left by the UAV. The process of reinforcement learning will be closed-loop since it will be among the UAV, the environment, and the RL agent

(Fig. 2). The environment provides the state with information about user locations, channel status (LoS/NLoS), interference, and energy of the UAV. Under this state, action selection will be done by the RL agent, and the trajectory optimization module will change the 3D position of the UAV based on the movement or hovering decision. This is then followed by the UAV reporting to the ground users to build on a new state, and a reward is offered based on communication performance and consumption of energy. A reward of this kind is used to amend the policy. Connection to the GBS is ensured by a backhaul connection. The sustained interaction allows the RL agent to change to an efficient and energy-saving route in dynamic UAV-based communication environments.

4 Algorithm Used

A Single-Agent Double Deep Q-Network (DDQN) is used to maximize the UAV trajectory optimization problem. The Markov Decision Process (MDP) is defined in which the UAV engages with a dynamic wireless environment to learn an energy-efficient flight policy.

4.1 Proposed Single-UAV DDQN Algorithm

1. Initialize online and target network along with replay buffer.
2. Track UAV energy and position.
3. For each episode:
 - a. Observe initial state.
4. For each time step:
 - a. Take action based on ϵ -greedy policy.
 - b. Execute action and change state.
 - c. Calculate reward and store transition.
 - d. Train online network using minibatch.
5. Periodically update target network.
6. Repeat until convergence.

5 Performance Evaluation Parameters

5.1 Achievable Data Rate

The instantaneous achievable data rate between the UAV and the ground user k at time step t is computed using Shannon's capacity formula:

$$R = B \log_2(1 + \text{SNR}_k(t)) \quad (12)$$

where B denotes the system bandwidth and $\text{SNR}_k(t)$ represents the signal-to-noise ratio of the user k .

5.2 Coverage Probability

Coverage probability is defined as the likelihood that a user experiences an SNR above a predefined threshold γ_{th} :

$$P_{cov} = Pr(\text{SNR} > \gamma_{th}) \quad (13)$$

5.3 Energy Efficiency

Energy efficiency is defined as the ratio of total transmitted data to total energy consumption

$$\eta = \frac{\sum_{t=1}^T \sum_{k=1}^K R_k(t)}{E_{\text{total}}} \quad (14)$$

5.4 Jain's Fairness Index

To evaluate fairness in resource allocation among users, Jain's fairness index is used:

$$J = \frac{(\sum_{k=1}^K R_k)^2}{K \sum_{k=1}^K R_k^2} \quad (15)$$

6 Result and Analysis

The intelligent UAV path is dynamically adjusted to the spatial distribution of the ground users. The RL-based trajectory, in contrast to the static one, which uses a defined route without considering where the users are, is redirected to areas with a greater number of users and good channels. This dynamic flight will enhance continuity of coverage and reduce redundant flight paths, leading to even better use of UAV energy. The convergence of cumulative reward over episodes is stable during training in the RL algorithm. First, performance improves because of exploration, but as learning increases, the UAV approaches a steady path [8] policy that can maintain a high data rate at reduced energy expenditure. This proves the usefulness of the suggested learning model in the optimization of trajectories in dynamic environments.

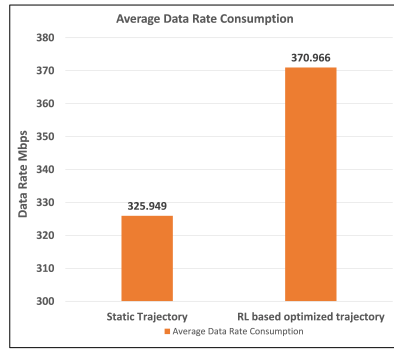


Fig. 3 Average Data Rate Consumption

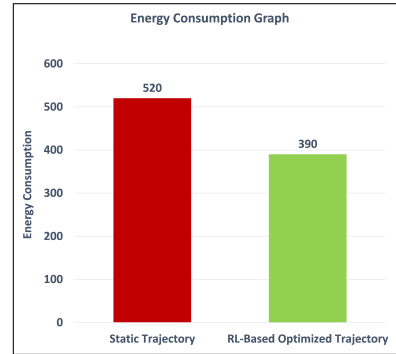


Fig. 4 Energy Consumption Graph

The data rate and energy consumption of the performance of the UAV based on static and RL trajectories are compared in both Fig.3 and Fig.4 . In Fig.3, the optimized path produces a better average data rate, improving it from 325 Mbps to 375 Mbps, In Fig.4, It is seen that the Energy Consumption of UAV got reduced to 490 from 510 which shows how optimized trajectory make UAV more energy efficient.

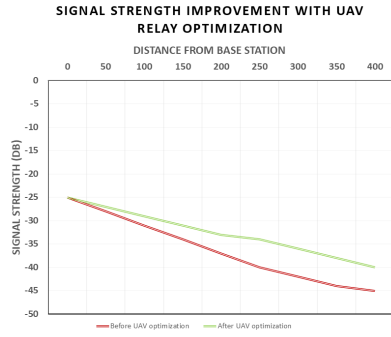


Fig. 5 Signal Strength Improvement

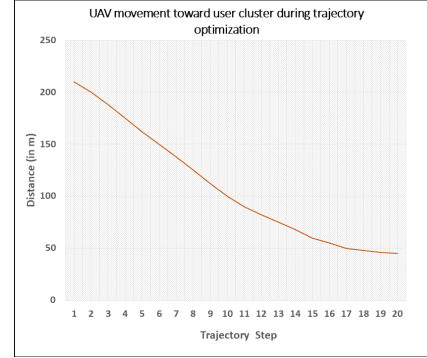


Fig. 6 UAV Distance from User Cluster

It is illustrated in Fig.5 that the optimized approach offers a higher signal strength at all distances, which points out to a lower path loss and the reliability of the links. It has been shown in Fig.6 how the UAV movement optimizes as the distance to the user cluster reduces with the increase in the number of steps in the trajectory, which ultimately proves to have successfully learnt user-centric positioning. In Fig.5, it can be clearly seen that the signal strength for users far from the base station is improved from -45 dB to -40 dB and in Fig.6, it can be seen that UAV is moving towards the user cluster and distance between UAV and user cluster is decreasing indicating that UAV is prioritizing crowded user area which in real world scenario can be a crowded concert, at the 20th trajectory step the distance of UAV from user cluster is less than 50m.

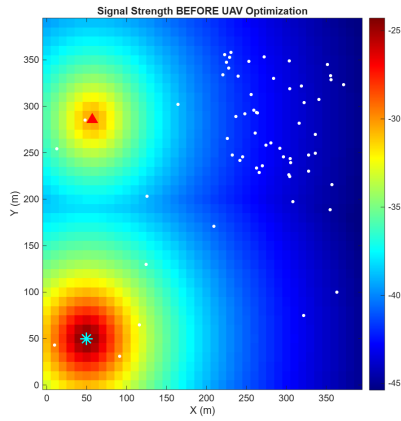


Fig. 7 Signal Strength Before Optimization

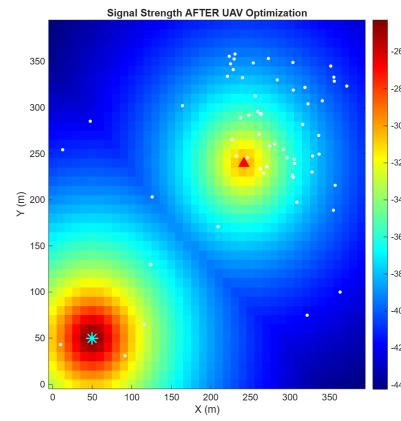


Fig. 8 Signal Strength After Optimization

The pre and post-optimization spatial signal strength distribution is represented by Fig.7 and Fig.8, respectively. Before optimization, there is unequal coverage with

the strong signals concentrated on the regions around the UAV, and most users have weak Signals in the range of -45 dB to -40 dB. Following the process of DDQN-based optimization, the UAV will move nearer to the user cluster while maintaining a backhaul with the base station, which will lead to an increased uniformity of signal distribution, fewer coverage holes, and the enhancement of communication performance in general.

7 Conclusion and Future Work

This paper provides a reinforcement learning-based algorithm to optimize UAVs' flight paths in 5G networks through a Double Deep Q-Network (DDQN). It is a model that includes the air-to-ground channel properties, energy limitations to provide effective mobility of the UAV, and a reliable communication channel. Among the metrics employed in the evaluation of performance are average achievable data rate, signal strength, coverage probability, fairness index, and UAV energy consumption. According to the simulation, optimization of the UAV relay enhances the average user signal strength of -40.8 dB to -32.6 dB (8.2 dB gain). The probability of coverage goes up to 0.81 compared to 0.46, and the fairness index goes up to 0.88 compared to 0.67, which means that there is a more balanced distribution of signals among numerous users. The strategic learning process also shows a stable convergence, which confirms the efficiency of DDQN. The results imply the power of reinforcement learning in supporting adaptive and energy-efficient UAV-assisted 5G deployments, especially in emergency and temporary network cases.

The proposed system can be extended to a multi-UAV cooperative communication framework to achieve wider and more efficient network coverage. The simulation environment can also be extended to support real-time wireless network simulation with dynamically moving random users to better simulate real-world scenarios. In addition, obstacles and urban building structures can be added for realistic UAV trajectory planning, and advanced fading and channel models can be used to simulate practical wireless propagation conditions in a more realistic manner.

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